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1.

The mean or expected value of a scalar random variable is a special case of the expression for the mean of a vector random variable. If a scalar random variable has pdf $f_X(x)$, then its mean is computed as $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$.

For this problem, consider a scalar random variable X having the following pdf:

$$f_X(x) = \begin{cases} 0.5 & 0 \leq x < 2 \\ 0 & \text{otherwise,} \end{cases}$$

What is $\mathbb{E}[X]$? Enter your answer with one digit to the right of the decimal point.

1

Correct

Yes. That is the correct calculation.

1 / 1 point
2.

If two jointly distributed random variables are uncorrelated, are they independent also?

☐ Yes: they are always independent also.

☒ Sometimes: they may be independent or they may not be independent.

☐ No: they are never independent also.

Correct

Yes. This is correct.

1 / 1 point
3.

Which of the following statements are always true about Gaussian random processes. Select all that apply.

☒ The pdf of the random process at any point in time can be completely specified by the mean and covariance of the process at that point in time.

Correct

Yes. This is correct.

☐ Gaussian random processes are always white.

☐ Gaussian random processes are always stationary.

☒ They can be reasonable models of noises that are themselves caused by the summation of many independent random causes.

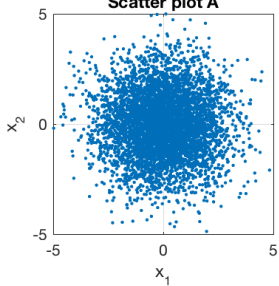
Correct

Yes. This is correct (by the Central Limit Theorem).

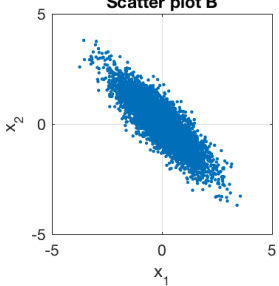
1 / 1 point
4.

Consider the four scatter plots shown below. Each plot displays 5000 samples of a Gaussian random variable having zero mean and an unknown covariance.

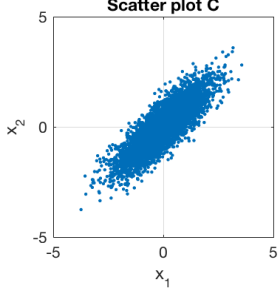
Scatter plot A



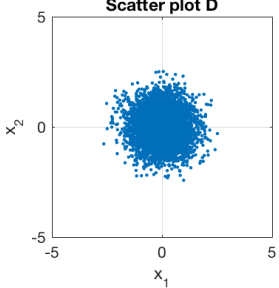
Scatter plot B



Scatter plot C



Scatter plot D



Which one of these plots could have been created from sampling a Gaussian random vector having covariance matrix $\Sigma = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$?

☐ Scatter plot A

☐ Scatter plot B

☐ Scatter plot C

☒ Scatter plot D

Correct

Yes. This is the correct answer. Notice that x_1 and x_2 are uncorrelated and that the 3σ standard deviation of each is on the order of 2.0.

1 / 1 point
5.

Consider a discrete-time system having $A_d = 0.9$, $B_d = 0.2$, $\hat{x}_0 = -1$, $u_0 = 5$, $\Sigma_{d,0} = 2$, and $\Sigma_w = 1$. What is $\Sigma_{d,1}$?

2.62

Correct

Yes. That is correct.

1 / 1 point
6.

Consider a discrete-time system having $A_d = 0.9$, $B_d = 0.2$, $\hat{x}_0 = -1$, $u_0 = 5$, $\Sigma_{d,0} = 2$, and $\Sigma_w = 1$. What is the steady-state value of Σ_d ? Enter your answer with two digits to the right of the decimal point.

5.263158

Correct

Yes. This is correct.

1 / 1 point
7.

Consider a continuous-time system having $A = -0.2$, $B = 0.5$, $B_w = 1$, $\hat{x}(0) = 10$, $u(0) = 2$, $\Sigma_d(0) = 2$, and $S_w = 1$. What is $\hat{x}(0)$?

-1

Correct

Yes. This is correct.

1 / 1 point
8.

Consider a continuous-time system having $A = -0.2$, $B = 0.5$, $B_w = 1$, $\hat{x}(0) = 10$, $u(0) = 2$, $\Sigma_d(0) = 2$, and $S_w = 1$. What is the steady-state value of Σ_d ?

2.5

Correct

Yes. That is correct.

1 / 1 point
9.

Consider a continuous-time system having $A = -0.2$, $B = 0.5$, $C = 3$, $D = 0$, $B_w = 1$, $S_w = 1$, and $S_v = 0.02$. If the sample period is $\Delta t = 0.1$, what is Σ_d ? You may use the Coursera Labs Jupyter notebook from Lesson 1.3.7 to compute the answer to this question. Enter your result with three digits to the right of the decimal point.

0.0980264

Correct

Yes. This is correct.

1 / 1 point
10.

Consider a continuous-time system having $A = -0.2$, $B = 0.5$, $C = 3$, $D = 0$, $B_w = 1$, $S_w = 1$, and $S_v = 0.02$. If the sample period is $\Delta t = 0.1$, what is Σ_d ? You may use the Coursera Labs Jupyter notebook to compute the answer to this question. Enter your result with two digits to the right of the decimal point.

0.2

Correct

Yes. That is correct.

1 / 1 point